

Vision-based UAV Detection Models for Small-Edge Devices*

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This research explores the development of **computer vision systems** designed to identify (**UAVs**) in real time, particularly for use on **edge devices** with limited processing power. The authors propose using **deep learning models** trained on a combination of public and custom-made datasets. The study evaluates several state-of-the-art algorithms (SOTA), including **YOLOv10** and **RT-DETR**, comparing their **detection accuracy** against their **operational speed**.

The authors identified that existing detection systems struggle with small object sizes and complex backgrounds, so they developed a **manually annotated dataset of approximately 2,000 images**. These images specifically capture "small target objects" with average area ratios of **0.013**, viewed from **ground-to-sky perspectives** to mimic real-world combat or security scenarios. And to improve robustness, they combined their manual data with several **publicly available datasets - Drone-vs-Bird** and **DUT-Anti-UAV** - which include challenges like motion blur, environmental disturbances, and various lighting conditions. They applied image augmentation, including **flipping, blurring, and color changes**, to help models generalize better to the variations found in real-world environments.

The authors evaluated and optimized several (**SOTA**) computer vision models to find the best balance between accuracy and speed. They tested single-stage (**YOLOv10**), two-stage (**Faster R-CNN**, **Mask R-CNN**), and transformer-based (**RT-DETR**) detectors.

For the **YOLO** family, they found that **freezing approximately 80% of the layers** and only fine-tuning the final detection layers yielded the best results. For **RT-DETR**, they utilized **full fine-tuning for 10 epochs** to leverage its transformer-based architecture for high-accuracy multi-scale feature processing.

The authors compared models not just on accuracy (**mAP@50**) but also on computational complexity (**GFLOPs**) and inference speed (**FPS**). They concluded that while **RT-DETR** offers the highest precision (0.971 mAP), lightweight **YOLO-based models** strike the best balance for real-time monitoring on low-power hardware due to their high throughput.

Their findings indicate that while **transformer-based models** offer superior precision, lightweight **YOLO variants** provide a more practical balance for real-world deployment on portable hardware. Ultimately, the paper demonstrates that modern **artificial intelligence** can effectively enhance situational awareness in military and security contexts by overcoming the limitations of conventional surveillance.